# Advanced EDA & Feature Engineering: Change Log

DecodeLabs Data Science | Project 1 | Batch 2026

Data cleaning, imputation, outliers and feature engineering · intern: Praise · notebook: project1_eda_feature_engineering.ipynb

## 1. Dataset Overview

| | |
|---|---|
| **Source file** | data/Dataset for Data Analytics.xlsx (one sheet, synthetic order data) |
| **Size** | 1,200 rows x 14 columns |
| **Columns** | OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice |
| **Date range** | 1 January 2023 to 30 June 2025 |
| **Target** | None supplied. **is_failed** was defined for this project: 497 orders (41.42%) were Cancelled or Returned. |
| **Tool** | Python (pandas, NumPy, scikit-learn) in Jupyter Notebook |
| **Deliverables** | project1_eda_feature_engineering.ipynb, data/cleaned_data.csv, data/engineered_data.csv, data/model_matrix.csv, this change log |

## 2. Change Log

One row per decision that altered the data or the way it is read. The number in *Impact* is measured, not estimated.

| ID | Description | Impact | Status |
|---|---|---|---|
| **CR001** | Loaded the workbook with pandas and kept the raw table as an untouched reference copy. Every change happens on a copy in memory, so the source file on disk is never overwritten and the starting point can always be re-read. | 1,200 rows x 14 columns loaded. Raw workbook unchanged. | Resolved |
| **CR002** | Audited every column for missing values and applied the brief's decision matrix (under 5% drop, 5-20% impute, over 20% KNN). **CouponCode** was the only column with gaps, at 25.75%, which the matrix sends to KNN. I did not follow it: the blanks sit alongside three coupon codes in almost equal shares, and a missing coupon is a fact about the order, not a lost measurement. Imputing would have invented a discount for orders that had none. | 309 blanks (25.75%) became an explicit <No Coupon> category. 0 records dropped, 0 coupons invented. | Resolved (assumption A1) |
| **CR003** | Converted **Date** from text to a real datetime and asserted that every value parsed. The export writes it back as ISO 8601 (YYYY-MM-DD) so the file is readable by any tool without a locale guess. | 0 unparseable dates. 0 wrongly formatted dates in the exported file, checked after reloading it from disk as text. | Resolved |
| **CR004** | Rounded the money columns to two decimals and trimmed leading and trailing spaces from every text column. Invisible whitespace in a category name silently creates a new category, which is the kind of fault that never raises an error. | Both applied to all 1,200 rows. TotalPrice = round(Quantity x UnitPrice, 2) then verified on every row. | Resolved |

| ID | Description | Impact | Status |
|---|---|---|---|
| **CR005** | Audited for duplicate records two ways: repeated **OrderID** values, and whole rows that are identical apart from the ID. | 0 duplicate OrderIDs, 0 fully duplicated rows. Nothing removed. | Resolved (no action needed) |
| **CR006** | Investigated the repeated **CustomerID**s rather than deleting them. A customer who orders three times is not three copies of one row, so the test is whether the rest of the record differs. | 11 customers appear more than once. 0 were hidden duplicates; 0 of them share a date, 0 share a tracking number and 0 share an address, so all rows were kept. | Resolved |
| **CR007** | Demonstrated the three imputation methods the brief asks for, on a throwaway copy. No numeric column had a gap, so there was nothing real to fill: I hid 10% of **UnitPrice** at random with seed 42, filled it with Mean, Median and KNN, and scored each against the values I had hidden. KNN was also run without **TotalPrice**, to see how much of its advantage came from that one column. | KNN with TotalPrice: RMSE 21.93. Mean 190.30, Median 190.45, KNN without TotalPrice 223.88. The real data was not modified. | Resolved (assumption A3) |
| **CR008** | Scanned for outliers with both rules the brief names, the IQR fence and $\|z\| > 3$. The flagged orders were not errors: they are five units of a high-priced product, so removing them would delete real sales. I capped them into a new column with **np.clip** and kept the original, so the price identity still holds. | 8 IQR outliers in TotalPrice, 0 by Z-score (largest $\|z\|$ is 2.93). 8 values capped into TotalPrice_capped. 0 rows lost. | Resolved |
| **CR009** | Built five new features, each as a whole-column pandas or NumPy operation rather than a row loop, as the brief asks: **used_coupon**, **other_items_in_cart**, **order_month**, **is_weekend** and **price_tier**. An assertion confirms the basket is never smaller than the quantity ordered. | 5 features added. 0 rows with a negative remainder in other_items_in_cart. | Resolved |
| **CR010** | Defined a target, because the dataset has none: **is_failed** is 1 when the order was Cancelled or Returned. It is used for one job only, breaking ties between correlated pairs. **OrderStatus** is then dropped from the features, because leaving it in would hand the model the answer. | 497 of 1,200 orders (41.42%) marked failed. OrderStatus absent from the model matrix. | Resolved (assumption A2) |
| **CR011** | One-hot encoded the five categorical columns with **drop_first=True**, so each has a baseline category instead of a full set of perfectly dependent dummies. Dropped six columns that cannot generalise: the three identifiers, the date (replaced by its features), the status (the target) and the shipping address. | 5 categorical columns became 20 dummy columns. Feature matrix 29 columns wide before collinearity removal. | Resolved |
| **CR012** | Removed collinear features with the brief's rule: take pairs correlated above 0.80, and in each pair drop whichever is less related to the target. Both flagged pairs turned out to be mirror images created by my own encoding, so nothing informative was lost. | 2 pairs above 0.80: used_coupon with CouponCode_No Coupon (1.000) and TotalPrice with TotalPrice_capped (0.99998). Dropped 2 columns; highest remaining pair 0.79. | Resolved |

| ID | Description | Impact | Status |
|---|---|---|---|
| CR013 | Exported two files and then checked the files, not the tables in memory: reloaded them from disk and re-asserted uniqueness, date format, no missing values, row count, the price identity and the absence of any remaining collinear pair. | engineered_data.csv 1,200 x 21; model_matrix.csv 1,200 x 28. All output checks passed. | Resolved |
| CR014 | Excluded **ShippingAddress** after checking it: the street name is *Main St* in every row, so the only variation is the house number, which identifies a household rather than describing an order. | 1 distinct street name in 1,200 rows. Column dropped rather than encoded into 1,200 meaningless dummies. | Resolved (deliberate) |

## 3. Before vs. After

| Metric | Raw input | Delivered | Comment |
|---|---|---|---|
| Rows | 1,200 | 1,200 | No row deleted at any step |
| Columns in the readable file | 14 | 21 | +5 features, +TotalPrice_capped, +is_failed |
| Columns in the model matrix | - | 28 | 27 features + target |
| Missing values in the data | 309 | 0 | Structural blanks given their own category |
| Values invented by imputation | 0 | 0 | The simulation ran on a copy |
| Values changed by capping | 0 | 8 | New column; the original kept |
| Duplicate records | 0 | 0 | None existed |
| Highest feature-to-feature \|corr\| | - | 0.79 | Below the 0.80 threshold |
| Strongest feature-to-target \|corr\| | - | 0.065 | PaymentMethod_Online |

> **The short version.** 1,200 orders in, 1,200 orders out. One column of blanks was given its own category instead of being filled in, eight extreme totals were capped into a separate column, five features were added and two mirror-image columns were removed. Nothing was deleted and nothing was invented.

## 4. The Three Brief Requirements, Measured

### 4.1 Missing data: the rule said KNN, and following it would have been wrong

Only **CouponCode** had gaps: 309 of 1,200 rows (25.75%). At that share the brief's matrix prescribes KNN. But the three coupon codes and the blanks each account for roughly a quarter of the orders, which is the shape of a deliberate "no coupon used" entry, not of lost data. Filling those 309 orders would have manufactured a discount in a quarter of the dataset. The blanks became a fourth category instead, and all 309 were kept.

Because nothing numerical was actually missing, the three methods were compared on a copy where 10% of **UnitPrice** (120 values) was hidden at random with seed 42, then recovered and scored against the values that had been hidden.

| Method | RMSE on the hidden values | Std of the filled column | Gap to the true std (197.18) | Reading |
|---|---|---|---|---|
| Mean | 190.30 | 187.76 | -9.42 | Carries no information about the missing row |
| Median | 190.45 | 187.78 | -9.40 | Carries no information about the missing row |
| KNN (all numeric columns) | 21.93 | 196.53 | -0.65 | Best, but only because it can see the answer |
| KNN (without TotalPrice) | 223.88 | 189.76 | -7.42 | Worst of the four |

Two things follow, and they matter more than the ranking. First, the winning method wins for a reason that would not survive contact with real data: **TotalPrice** is literally **Quantity x UnitPrice**, so it states the answer. Remove it and KNN (223.88) does worse than replacing every gap with the mean (190.30). Imputation quality depends on the other columns, not on the algorithm. Second, mean and median filling shrink the spread of the column from 197.18 to about 187.76: they flatten the distribution even when they look acceptable on a single number. Best RMSE in this run: 21.93.

### 4.2 Outliers: both rules, and why the rows were not deleted

| Column | IQR lower fence | IQR upper fence | Flagged by IQR | Flagged by \|z\| > 3 | Largest \|z\| |
|---|---|---|---|---|---|
| Quantity | -1.00 | 7.00 | 0 | 0 | 1.46 |
| UnitPrice | -317.20 | 1,024.83 | 0 | 0 | 1.75 |
| ItemsInCart | -0.50 | 11.50 | 0 | 0 | 1.98 |
| TotalPrice | -1,341.41 | 3,330.41 | 8 | 0 | 2.93 |

The Z-score rule flags nothing at all, because the largest standardised value in the whole table is 2.93. The IQR rule flags the 8 largest totals, and each of them is five units of a product priced near the top of its range: a real, expensive order, not a mis-keyed one. Deleting them would have removed actual revenue from the dataset, so they were winsorised into **TotalPrice_capped** and the original column was kept, which is what lets the price identity still be checked.

### 4.3 Feature engineering: five features, and an honest reading of them

| Feature | Built from | Why it might help | \|corr\| with is_failed |
|---|---|---|---|
| used_coupon | CouponCode | A discount can change whether an order completes | 0.023 |
| other_items_in_cart | ItemsInCart, Quantity | Separates a big basket from a big quantity of one product | 0.011 |
| order_month | Date | Captures seasonality | 0.055 |
| is_weekend | Date | Captures weekday and weekend behaviour | 0.040 |
| price_tier | UnitPrice | Groups prices into four bands: Low, Mid, High, Premium | 0.017 |

The last column is the uncomfortable one, and it belongs in the log rather than in a footnote. The strongest engineered feature, **order_month**, reaches only 0.055, and with 1,200 rows a correlation of about ±0.057 would arise by chance alone. So none of the five features can be shown to carry signal here. They are still the right features to build, because the brief asks for them and because each encodes something a business would plausibly act on, but the dataset is synthetic and nearly uniform, and no amount of feature work changes that. The strongest relationship anywhere in the matrix, PaymentMethod_Online 0.065, order_month 0.055, CouponCode_SAVE10 0.053, is also barely above the noise floor.

## 5. From 29 Columns to 27

Encoding produced 29 columns: 9 numeric and 20 dummies, with one category per column held back as the baseline. Two pairs then exceeded the 0.80 threshold.

| Column A | Column B | \|corr\| | Dropped | That column's \|corr\| with the target |
|---|---|---|---|---|
| used_coupon | CouponCode_No Coupon | 1.00000 | CouponCode_No Coupon | 0.0233 |
| TotalPrice | TotalPrice_capped | 0.99998 | TotalPrice | 0.0086 |

Both pairs are artefacts of my own construction, not discoveries: **used_coupon** is the exact complement of the **CouponCode_No Coupon** dummy, and **TotalPrice_capped** is **TotalPrice** with 8 values clipped. Each pair carries the same information twice, so keeping either side teaches a model the same thing. The rule dropped 2 columns and the highest remaining pair is 0.79 (**ItemsInCart** and **other_items_in_cart**, one derived from the other, deliberately left just under the line because both are wanted).

## 6. Verification Gate

These are checks on the delivered files, not on a description of them. Every figure in this document was recomputed from **data/** while the PDF was built; if the files and the notebook ever disagree, this document reports FAIL rather than repeating the notebook.

| Check | Requirement | Measured | Result |
|---|---|---|---|
| Exact duplicate rows | None in the raw workbook | 0 found, 0 duplicate OrderIDs | **PASS** |
| Every step kept all rows | 1,200 rows in, 1,200 rows out, at every stage | 1,200 raw, 1,200 cleaned, 1,200 in the model matrix | **PASS** |
| No imputation on real data | Nothing invented: the only blanks were structural | 0 missing values in cleaned_data.csv, 0 in model_matrix.csv | **PASS** |
| The simulation never touched the data | UnitPrice in the delivered file equals the raw one | identical | **PASS** |
| OrderIDs unique | No duplicated order identifier | 1,200 unique of 1,200 | **PASS** |
| Dates are ISO 8601 in the delivered file | YYYY-MM-DD on every row | 0 wrongly formatted | **PASS** |
| Arithmetic still holds | TotalPrice = round(Quantity x UnitPrice, 2) | holds on all rows | **PASS** |
| No collinear pair left | Every remaining pair <= 0.80 | highest remaining 0.788 | **PASS** |
| Target absent from the features | OrderStatus must not reach the model matrix | absent | **PASS** |
| Model matrix reproducible | Rebuilt from engineered_data.csv, column for column | identical | **PASS** |

| Notebook output, re-run from the committed files | Notebook | Rebuilt | Result |
|---|---|---|---|
| Hidden UnitPrice values | 120 | 120 | **PASS** |
| True std of UnitPrice | 197.18 | 197.18 | **PASS** |
| Rows flagged by IQR in TotalPrice | 8 | 8 | **PASS** |
| Values actually capped | 8 | 8 | **PASS** |
| Feature matrix before removal | 1,200 x 29 | 1,200 x 29 | **PASS** |
| Pairs above 0.80 | 2 | 2 | **PASS** |

| Notebook output, re-run from the committed files | Notebook | Rebuilt | Result |
|---|---|---|---|
| Feature matrix after removal | 1,200 x 27 | 1,200 x 27 | PASS |
| engineered_data.csv | 1,200 x 21 | 1,200 x 21 | PASS |
| model_matrix.csv | 1,200 x 28 | 1,200 x 28 | PASS |

| Imputation: hidden values recovered | RMSE (notebook) | RMSE (rebuilt) | Std (notebook) | Std (rebuilt) | Result |
|---|---|---|---|---|---|
| Mean | 190.30 | 190.30 | 187.76 | 187.76 | PASS |
| Median | 190.45 | 190.45 | 187.78 | 187.78 | PASS |
| KNN (all numeric cols) | 21.93 | 21.93 | 196.53 | 196.53 | PASS |
| KNN (without TotalPrice) | 223.88 | 223.88 | 189.76 | 189.76 | PASS |

The rebuild applies the notebook's own steps to the delivered CSVs: seed 42 for the hidden values, **KNNImputer** with 5 neighbours on scaled columns, the same IQR and Z-score rules, the same one-hot settings and the same collinearity rule. Tolerance 0.005 on the rounded figures. Overall: **every check passes**.

## 7. Notes & Assumptions

- **A1.** A blank **CouponCode** means no coupon was used. This overrides the brief's missing-value matrix on purpose, and it is the single most consequential judgement in the project: it keeps 309 orders and adds no invented discounts. If the blanks were in fact lost data, the right answer would have been KNN instead.
- **A2.** The dataset has no target. **is_failed** = Cancelled or Returned is my definition, written down so it can be challenged. It is used only to decide which column of a collinear pair to drop, and it is *not* a validated label: no model is trained or scored on it here. Awaiting instructor confirmation.
- **A3.** The imputation comparison is a simulation on a throwaway copy. It demonstrates the methods the brief requires; it does not repair anything, because nothing needed repair. Awaiting instructor confirmation.
- **A4.** Outliers were capped rather than removed, following the brief's winsorisation guidance. Capping is not neutral: it compresses the top of the price distribution, and a later analysis of revenue would need the original column back. That is why both columns are exported.
- **A5.** The 0.80 collinearity threshold is a rule of thumb. The closest surviving pair sits at 0.79, and a threshold of 0.75 would have flagged 2 further pairs. Nothing in this project depends on that choice.
- **A6.** The brief recommends **Pandera** for schema contracts, and it is listed in requirements.txt, but the notebook does not use it. Validation is a set of explicit assertions on the reloaded files (unique OrderIDs, ISO dates, no nulls, the price identity, no remaining collinear pair). Those assertions test values this project cares about, which a type-and-range schema would not catch. This is a deviation from the recommendation, stated here rather than described as done.

## 8. Limits, Reproducibility and Attribution

- **Limits.** The data is synthetic, with near-uniform numeric columns and evenly split categories, so the weak correlations reported above are a property of the data, not a failure of the processing. No model was trained, so nothing here demonstrates that these five features improve a prediction; that evidence would have to come from a model evaluated on held-out data. What this project can show is that the pipeline is correct, documented and reproducible.
- **Reproducibility.** Rebuilt with Python 3.13.9, pandas 2.3.3, NumPy 2.3.5 and scikit-learn 1.7.2. Paths are relative to the notebook, so it must stay beside **data/**. Kernel → Restart & Run All regenerates all three CSVs and every number in this document.
- **Attribution.** Dataset supplied with the DecodeLabs brief. Synthetic, containing no real customers; the identifiers and addresses are generated.